

24 The siren on an emergency vehicle emits a sound of constant frequency 1100 Hz.

The driver in a stationary car hears the sound at a frequency of 1000 Hz.

The speed of sound in air is 340 m s^{-1} .

What is the velocity of the emergency vehicle?

- A** 31 m s^{-1} away from the stationary car
- B** 34 m s^{-1} away from the stationary car
- C** 31 m s^{-1} towards the stationary car
- D** 34 m s^{-1} towards the stationary car